

Project Abstract: Multi-Paradigm Machine Learning for Sentiment Analysis on the IMDB Movie Review Dataset

Anjali Kumari, Paripelly Pritika, Sucharitha Kamera, Harshith Konda, and
Tangirala Dhanunjaya Rao

Data Speaks!! Course Assignment

Project Overview

The “Data Speaks!!” project explores the capacity of machine learning architectures to decode human sentiment and predict exact user ratings from unstructured natural language text. Utilizing the Large Movie Review Dataset (IMDB), the project tests the limitations of traditional statistical models against modern Deep Learning mechanisms across multiple learning paradigms.

Dataset Engineering and EDA

To satisfy the multi-paradigm requirements, 100,000 raw text files were systematically parsed into three structured datasets:

- **Binary Classification (50,000 reviews):** Extracted from explicitly labeled positive and negative folders, creating a perfectly balanced dataset.
- **Continuous Regression (50,000 reviews):** Derived by extracting the exact 1–10 numerical ratings embedded within the raw filename metadata.
- **Unsupervised (50,000 reviews):** Extracted from unlabelled directories to serve as a blind clustering sandbox.

Extensive Exploratory Data Analysis (EDA) revealed key meta-features:

- Bigrams (e.g., “highly recommend”) were exponentially more discriminative than generic unigrams (stopwords).
- A U-shaped distribution in review length indicated that polarized opinions (1-star and 10-star) motivate significantly longer reviews than moderate opinions.
- Punctuation density (exclamation marks, capitalization) heavily correlated with 1-star negative sentiment.

Methodologies and Key Results

Binary Classification: The project evaluated the predictive power of four models on the binary positive/negative task. Naive Bayes achieved 84.20% accuracy, and Logistic Regression achieved 88.22%. To capture sequential language dependencies, a Bidirectional LSTM was implemented (87.78%). Ultimately, by leveraging bidirectional self-attention, a fine-tuned BERT model achieved state-of-the-art **94.16% accuracy**.

Continuous Regression: Mapping text to an exact 1–10 numerical rating was evaluated using three models adapted with Mean Squared Error (MSE) objective functions. An XG-Boost model on TF-IDF features yielded a Mean Absolute Error (MAE) of 2.1612. An LSTM model improved this to 1.5477. However, the BERT Regression model significantly outperformed both, achieving an MAE of **0.9897** ($R^2 = 0.8154$). This implies BERT can predict a user’s exact rating within roughly one single point based solely on the textual review.

Dimensionality Reduction and Clustering: To fulfill unsupervised requirements, Truncated SVD was used to compress 5,000-dimensional sparse TF-IDF vectors into 2 Principal Components, validating spatial separation between positive and negative linguistic profiles. Two distinct K-Means clustering approaches (including the dedicated `Sentiment_Analysis_KMeans1.ipynb` pipeline) were explored to group the 50,000 unlabelled reviews topically, minimizing within-cluster spatial distances to discover natural latent sentiment groupings. Gaussian Mixture Models (GMM) were then applied as a probabilistic alternative, identifying overlapping soft-cluster sentiment profiles.

Summary

Contextual understanding (BERT) drastically outperforms bag-of-words approaches (TF-IDF) across both classification and regression paradigms. Furthermore, analyzing user writing style (length, punctuation) provides immense secondary predictive value. All analytical findings and interactive models were ultimately integrated into a responsive web application to disseminate these insights.